The empirical recurrence for OEIS A263871, recovered from its tail

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Abstract

OEIS A263871 records "Empirical recurrence of order 79" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4168$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 80$ — every index at which it can be written down. Its last coefficient is $c_{79} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-79 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A263871 is "Number of (n+1)X(6+1) 0..1 arrays with each row and column divisible by 3, read as a binary number with top and left being the most significant bits, and rows and columns lexicographically nondecreasing.". It has offset 1 and begins

$$4, 4, 14, 17, 93, 379, 2909, 20374, \ \ldots$$

The entry states:

> Empirical recurrence of order 79 (see link above)

As of the "Last modified" line on the live entry (Oct 28 2015 11:49:07, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4168$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4168$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8336$ exact terms from $n = 1$ on, it returns order 79 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{79} c_i\, a(n-i),$$

c_{1}	1	c_{21}	635745396	c_{41}	68923264410	c_{61}	211915132
c_{2}	39	c_{22}	1676056044	c_{42}	62359143990	c_{62}	61523748
c_{3}	−39	c_{23}	−1676056044	c_{43}	−62359143990	c_{63}	−61523748
c_{4}	−741	c_{24}	−3910797436	c_{44}	−51021117810	c_{64}	−15380937
c_{5}	741	c_{25}	3910797436	c_{45}	51021117810	c_{65}	15380937
c_{6}	9139	c_{26}	8122425444	c_{46}	37711260990	c_{66}	3262623
c_{7}	−9139	c_{27}	−8122425444	c_{47}	−37711260990	c_{67}	−3262623
c_{8}	−82251	c_{28}	−15084504396	c_{48}	−25140840660	c_{68}	−575757
c_{9}	82251	c_{29}	15084504396	c_{49}	25140840660	c_{69}	575757
c_{10}	575757	c_{30}	25140840660	c_{50}	15084504396	c_{70}	82251
c_{11}	−575757	c_{31}	−25140840660	c_{51}	−15084504396	c_{71}	−82251
c_{12}	−3262623	c_{32}	−37711260990	c_{52}	−8122425444	c_{72}	−9139
c_{13}	3262623	c_{33}	37711260990	c_{53}	8122425444	c_{73}	9139
c_{14}	15380937	c_{34}	51021117810	c_{54}	3910797436	c_{74}	741
c_{15}	−15380937	c_{35}	−51021117810	c_{55}	−3910797436	c_{75}	−741
c_{16}	−61523748	c_{36}	−62359143990	c_{56}	−1676056044	c_{76}	−39
c_{17}	61523748	c_{37}	62359143990	c_{57}	1676056044	c_{77}	39
c_{18}	211915132	c_{38}	68923264410	c_{58}	635745396	c_{78}	1
c_{19}	−211915132	c_{39}	−68923264410	c_{59}	−635745396	c_{79}	−1
c_{20}	−635745396	c_{40}	−68923264410	c_{60}	−211915132		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 80$, and it is the recurrence OEIS A263871 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{79} - c_1 x^{79-1} - \cdots - c_{79}$. Its constant term is $-c_{79} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 79 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 79, so $\deg q = 79$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 79, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 79, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-79 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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